11.1 DirectX 10

DX10 is a set of APIs (an API is a layer of software that tells other software and hardware how to interact with it) that enables next-generation gaming; by "next-generation," we mean beyond what consoles would be capable of in the years to come. DirectX 10 will offer a variety of new features and new tricks to old tasks which will take the visual fidelity and the performance of games built using the API to a new level. With DX10 in particular and Vista in general, Microsoft has shifted the onus of graphics rendering from the processor to the graphics card.

To achieve that end, DX10 has been built from the ground up to change the way 3D applications think about material management and load balance between the CPU and GPU. Direct3D 10, the component of DX10 that manages the 3D rendering tasks, takes advantage of the improved communication between the CPU and GPU and efficiently manages the data transfer between them. Through its advanced material management and load balancing tricks, DX10 games will bring in scenes with far greater complexity than currently possible, without ever increasing the CPU overhead. This frees up the processor to do other tasks such as AI and physics calculations, which further increases the immersion factor of these games.

As has been mentioned in Chapter 2, DX10 brings along a new driver model for the graphic cards. The new display driver model called WDDM is the main reason for DirectX10's exclusivity to the operating system: Windows Vista will be the sole operating system (or at least the first) under which DirectX 10 and the games that support it will run. DirectX10 uses and leverages the virtualisation and architectural improvements of WDDM, in both the APIs as well as the underlying infrastructure.

11.1.1 In Greater Detail

DX10 allows programmers to create scenes with greater detail than before. For example, a technical demo of the *Crysis* game